SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct
I Santhosh kumar________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Santhosh

Assessment Tasks
Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.
2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.
   Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.
3. **Case Study 3: Online Food Ordering Platform**

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.
Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.
Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

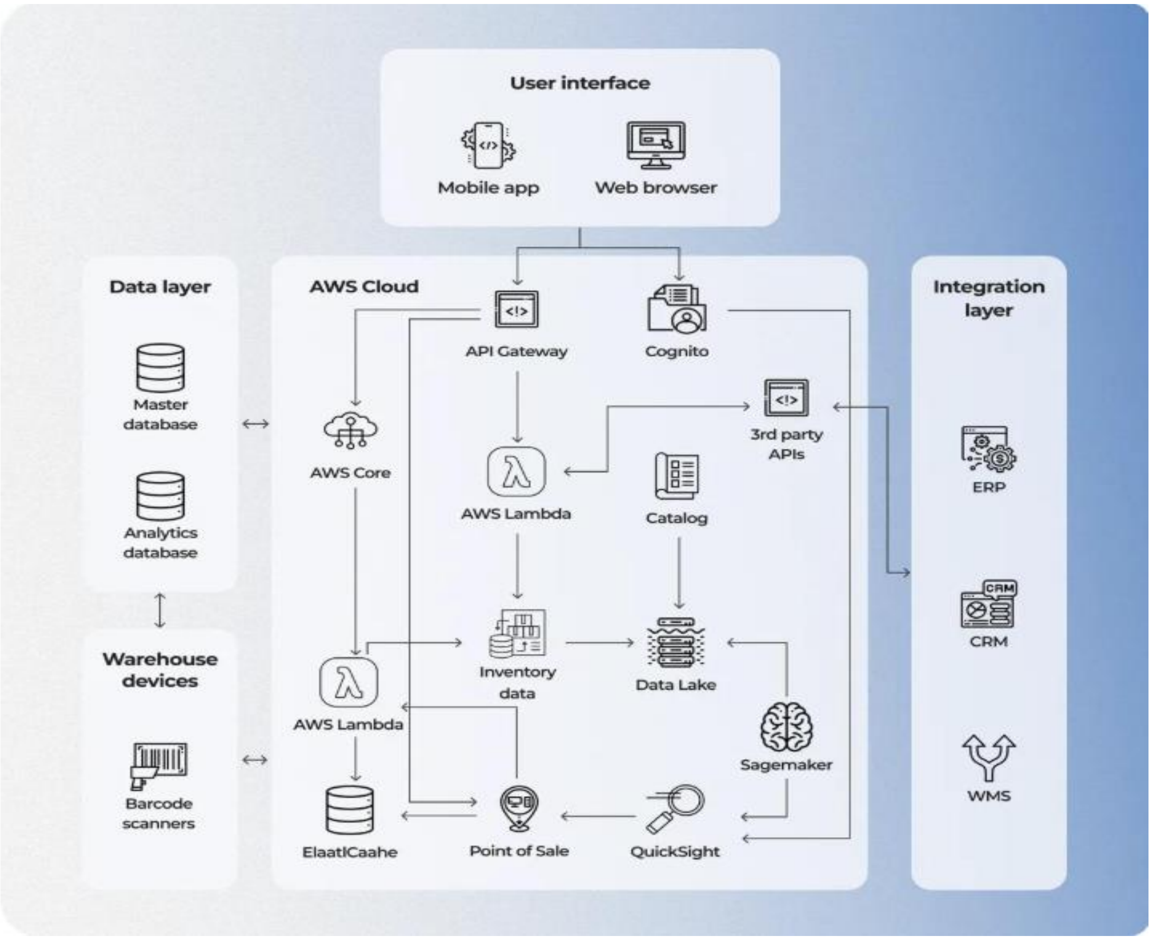
through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

Case Study 1: Smart Retail Inventory Management System

Architecture Diagram:



Key Components

- Clients: POS systems, inventory dashboards, and store devices.
- Web APIs: Enable communication between clients and cloud services.
- Cloud Storage: Stores product details, sales transactions, logs, and analytics data.
- Platform Services (PaaS): Used for demand forecasting and analytics.
- Infrastructure (IaaS): Virtual servers, networking, and load balancers.

Working

- POS systems send transaction data through APIs.
- Cloud services update inventory in real time.
- Analytics services process sales patterns.
- Forecasting predicts future demand and stock requirements.

Security

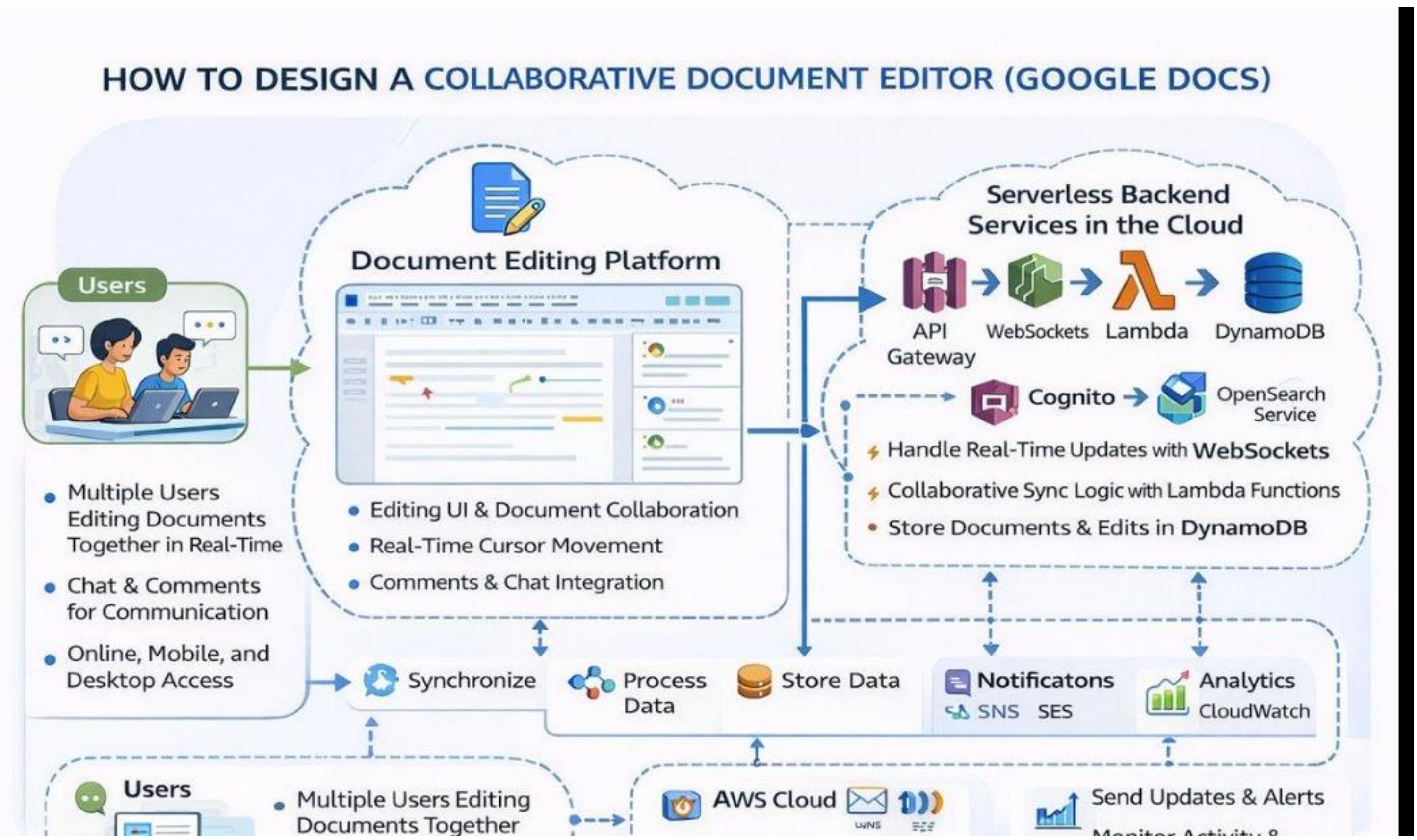
- VPN for secure store-to-cloud connectivity.
- Firewalls to filter network traffic.
- API Gateway for authentication, throttling, and monitoring.

Benefits

- Real-time inventory visibility.
- Reduced stock shortages and overstocking.
- Centralized management across all stores.
- Scalable analytics for business growth.

Case Study 2: Cloud-Based Document Collaboration System

Architecture Diagram:



Key Components

- Clients: Web browsers and mobile devices.
- Web APIs: Synchronize edits across users.
- Distributed Cloud Storage: Stores documents redundantly.
- Version Control: Maintains document history and recovery.
- Global Network Infrastructure: Provides low latency and high availability.

Working

- User edits a document in the browser.
- Changes are sent through APIs.
- Collaboration service merges updates in real time.
- Updated document is stored with a new version.

Security

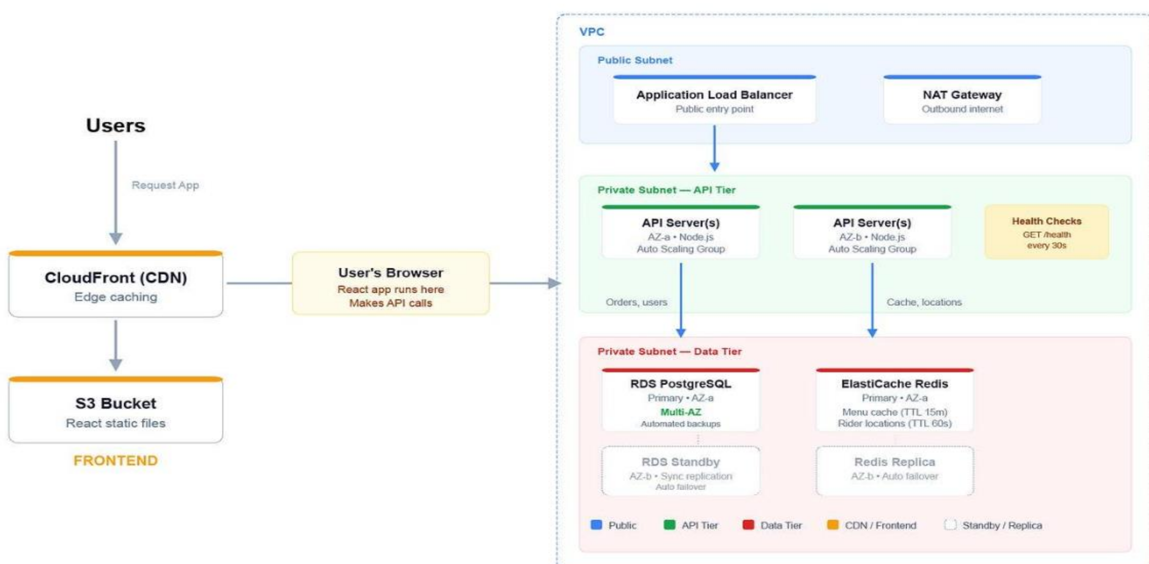
- Encryption in transit (HTTPS/TLS).
- Encryption at rest.
- Identity and access management.
- Role-based sharing permissions.

Benefits

- Real-time collaboration.
- Automatic backup and recovery.
- Access from any device.
- High availability across regions.

Case Study 3: Online Food Ordering Platform

Architecture Diagram :



Key Components

- Clients: Mobile apps and web browsers.
- RESTful APIs: Connect frontend and backend services.
- Cloud Storage: Stores menu images, invoices, and customer data.
- Virtual Machines (IaaS): Run supporting services.
- Managed Kubernetes: Handles container orchestration and scaling.

Working

- User places an order through the app.
- API sends the request to backend services.
- Order, payment, and restaurant services process the request.
- Data is stored securely in cloud storage/databases.

Security

- Authentication tokens (JWT/OAuth).
- HTTPS for secure communication.
- Role-Based Access Control (RBAC).
- Secure storage of customer information.

Benefits

- Handles traffic spikes during meal times.
- Fast deployment using containers.
- Reliable and scalable architecture.
- Secure real-time operations.